Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The materials suitable for making electromagnets should have

(A) high retentivity and low coercivity

(B) low retentivity and low coercivity

(C) high retentivity and high coercivity

(D) low retentivity and high coercivity

Q.2

If the susceptibility of dia, para and ferro magnetic materials are $\chi_{\mathrm{d}}, \chi_{\mathrm{p}}, \chi_{\mathrm{f}}$ respectively, then

(A) $x_{\mathrm{d}} < \chi_{\mathrm{p}} < \chi_{\mathrm{f}}$

(B) $\chi_{\mathrm{d}} < \chi_{\mathrm{f}} < \chi_{\mathrm{p}}$

(C) $\chi_{\mathrm{f}} < \chi_{\mathrm{d}} < \chi_{\mathrm{p}}$

(D) $x_f < \chi_p < \chi_d$

Q.3

Needles N_1, N_2 and N_3 are made of a ferromagnetic, a paramagnetic and a diamagnetic substance respectively. A magnet when brought close to them will

(A) attract N_1 and N_2 strongly but repel N_3

(B) attract N_1 strongly, N_2 weakly and repel N_3 weakly

(C) attract N_1 strongly, but repel N_2 and N_3 weakly

(D) attract all three of them

Q.4

A watch glass containing some powdered substance is placed between the pole pieces of a magnet. Deep concavity is observed at the centre. The substance in the watch glass is

(A) iron

(B) chromium

(C) carbon

(D) wood

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Q.5

A frog can be deviated in a magnetic field produced by a current in a vertical solenoid placed below the frog. This is possible because the body of the frog behaves as

(A) paramagnetic

(B) diamagnetic

(C) ferromagnetic

(D) antiferromagnetic

Q.6

A copper tod is suspended in a non-homogeneous magnetic field region. The tod when in equilibrum will align itself

(A) in the region where magnetic field is strongest

(B) in the region where magnetic field is weakest and parallel to direction of magnetic field there

(C) in the direction in which it was originally suspended

(D) in the region where magnetic field is weakest and perpendicular to the direction of magnetic field there

Q.7

The mass of a specimen of a ferromagnetic material is $0.6\ \mathrm{kg}$. and its density is $7.8 \times 10^3\ \mathrm{kg/m^3}$. If the area of hysteresis loop of alternating magnetising field of frequency $50\ \mathrm{Hz}$ is $0.722\mathrm{MKS}$ units then the hysteresis loss per second will be

(A) 277.7×10^{-5} Joule

(B) 277.7×10^{-6} Joule

(C) 277.7×10^{-4} Joule

(D) None of these

Q.8

A magnetic needle lying parallel to a magnetic field requires W units of work to turn it through 60°. The torque required to keep the needle in this position will be

(A) $2\ \mathrm{W}$

(B) W

(C) $\frac{\mathrm{W}}{\sqrt{2}}$

(D) $\sqrt{3}\ \mathrm{W}$

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Q.9

At a place, if the earth's horizontal and vertical components of magnetic fields are equal, then the angle of dip will be

(A) 30°

(B) 90°

(C) 45°

(D) 0°

Q.10

A dip needle lies initially in the magnetic meridian when it shows an angle of $\text{dip}\,\theta$ at a place. The dip circle is rotated through an angle x in the horizontal plane and then it shows an angle of $\text{dip}\,\theta'$. Then $\frac{\tan\theta'}{\tan\theta}$ is

(A) $\frac{1}{\cos x}$

(B) $\frac{1}{\sin x}$

(C) $\frac{1}{\tan x}$

(D) cos x

Q.11

A dip circle is so set that its needle moves freely in the magnetic meridian. In this position, the angle of dip is 40°. Now the dip circle is rotated so that the plane in which the needle moves makes an angle of 30° with the magnetic meridian. In this position, the needle will dip by an angle

(A) 40°

(B) 30°

(C) more than 40°

(D) less than 40°

Q.12

A compass needle placed at a distance r from a short magnet in Tan A position shows a deflection of 60°. If the distance is increased to $r(3)^{13}$, then deflection of compass needle is

(A) 30°

(B) $60 \times 3^{\frac{1}{3}}$

(C) $60 \times 3^{\frac{2}{3}}$

(D) $60 \times 3^{\frac{3}{3}}$

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Q.13

The radius of the coil of a tangent galvanometer is $16\ \mathrm{cm}$. How many turns of the wire should be used if a current of $40\ \mathrm{m\ A}$ is to produce a deflection of 45°. given, horizontal component of earth's field is $0.36 \times 10^{-4}\ \mathrm{T}$.

(A) 458

(B) 229

(C) 200

(D) 115

Q.14

A tangent galvanometer is connected directly to an ideal battery. If the number of turns in the coil is doubled, the deflection will

(A) increase

(B) decrease

(C) remain same

(D) any of these

Q.15

The magnetic needle of a tangent galvanometer is deflected at an angle 30° due to a magnet. The horizontal component of earth's magnetic field $0.34 \times 10^{-4} T$ is along the plane of the coil. The magnetic intensity is

(A) $1.96 \times 10^{-4} T$

(B) $1.96 \times 10^{-5}\ \mathrm{T}$

(C) $1.96 \times 10^{4}\ \mathrm{T}$

(D) $1.96 \times 10^{5} T$

Answers & Solutions

Q.1 Answer:

low retentivity and low coercivity

Solution:

Electro magnet should be amenable to magnetisation and demagnetization

$\therefore$ retentivity and coercivity should be low.

Q.2 Answer:

$x_{\mathrm{d}} < \chi_{\mathrm{p}} < \chi_{\mathrm{f}}$

Solution:

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$\chi_{\mathrm{d}} < \chi_{\mathrm{p}} < \chi_{\mathrm{f}}$

For diamagnetic substance χ_{d} is small and negative $\left(10^{-5}\right)$

For paramagnetic substances χ_p is small and positive $\left(10^{-3} \text{ to } 10^{-5}\right)$

For ferromagnetic substanes χ_{f} is very large $\left(10^{3} \text{ to } 10^{5}\right)$

Q.3 Answer:

attract N_1 strongly, N_2 weakly and repel N_3 weakly

Solution:

Ferromagnetic substance has magnetic domains whereas paramagnetic substances have magnetic dipoles which get attracted to a magnetic field. Diamagnetic substances do not have magnetic dipole but in the presence of external magnetic field due to their orbital motion of electrons these substances are repelled.

Q.4 Answer:

iron

Solution:

Iron is ferromagnetic.

Q.5 Answer:

diamagnetic

Solution:

Diamagnetic substances are repelled by the magnetic field.

Q.6 Answer:

in the region where magnetic field is weakest and perpendicular to the direction of magnetic field there

Solution:

As copper is a diamagnetic material, therefore copper rod will align itself in the region where magnetic field is weakest and pexpendicular to the direction of magpetic field.

Q.7 Answer:

277.7×10^{-5} Joule

Solution:

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$$W_H = VAft$$
$$= \frac{m}{d} Aft$$
$$\text{or } W_m = \frac{0.6}{7.8 \times 10^3} \times 0.722 \times 50$$
$$= 277.7 \times 10^{-5} \text{Joule}$$

Q.8 Answer:

$\sqrt{3}$ W

Solution:

Work done, $W = MB(\cos\theta_1 - \cos\theta_2)$
Here $\theta_1 = 0^\circ, \theta_2 = 60^\circ$
$\therefore \quad W = MB\left[1 - \frac{1}{2}\right] = \frac{MB}{2}$
Torque, $\tau = MB\sin\theta = MB\sin 60^\circ$
$= \frac{MB\sqrt{3}}{2} = \sqrt{3}$ W **(Using eq. (i))**

Q.9 Answer:

45°

Solution:

$$\tan\theta = \frac{B_V}{B_H} = 1 \quad \therefore \quad \theta = 45^\circ$$

Q.10 Answer:

$\frac{1}{\cos x}$

Solution:

$$\tan\theta = \frac{B_V}{B_H}; \tan\theta' = \frac{B_V}{B_H \cos x}; \frac{\tan\theta'}{\tan\theta} = \frac{1}{\cos x}$$

Q.11 Answer:

less than 40°

Solution:

$\delta_1 = 40^\circ, \delta_2 = 30^\circ, \delta = ?$
$\cot\delta = \sqrt{\cot^2\delta_1 + \cot^2\delta_2}$
$= \sqrt{\cot^2 40^\circ + \cot^2 30^\circ}$
$\cot\delta = \sqrt{1.19^2 + 3} = 2.1$
$\therefore \delta = 25^\circ$ i.e. $\delta < 40^\circ$.

Q.12 Answer:

30°

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Solution:

$\frac{\tan\theta_2}{\tan\theta_1} = \frac{d_1^3}{d_2^3} = \frac{r^3}{\left[r(3)^{1/3}\right]^3} = \frac{1}{3}$

$\tan\theta_2 = \frac{1}{3}\tan\theta_1 = \frac{\tan 60}{3} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$

$\therefore \quad \theta_2 = 30^\circ$

Q.13 Answer:

229

Solution:

No solution available

Q.14 Answer:

remain same

Solution:

$i = \left(\frac{2RB_H}{\mu_0 N}\right)\tan\theta$

or $\frac{V}{R} = \left(\frac{2RB_H}{\mu_0 N}\right)\tan\theta$ **...(i)**

When number of turns are doubled, resistance of the coil is also doubled, so

$\frac{V}{(2R)} = \left[\frac{2RB_H}{\mu_0(2N)}\right]\tan\theta'$ **. . . (ii)**

From (i) and (ii), $\theta' = \theta$**.**

Q.15 Answer:

1.96×10^{-5} T

Solution:

We know that $\frac{B}{B_H} = \tan\theta$

or $B = B_H \tan\theta$

$= 0.34 \times 10^{-4}\tan 30^\circ = 1.96 \times 10^{-5}$ T

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